

Practice Quiz 06A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 06

Existence and uniqueness as two separate claims.

Label and prove both halves.

1. Prove that for every real number $a \neq 0$, there is exactly one real number x such that $ax + 4 = 9$.

Solution Let $a \in \mathbf{R}$ be arbitrary and suppose $a \neq 0$.

Existence. Set $x = 5/a$. This is real because $a \neq 0$, and

$$ax + 4 = a \left(\frac{5}{a} \right) + 4 = 5 + 4 = 9.$$

Thus a solution exists.

Uniqueness. Suppose x and y both satisfy the equation. Then $ax + 4 = 9 = ay + 4$, so $a(x - y) = 0$. Since $a \neq 0$, it follows that $x - y = 0$, hence $x = y$. Thus the solution is unique. Since a was arbitrary, the claim holds for every nonzero real a .

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